# The minimum Kirchhoff index of polyomino chains

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**Abstract**

We determine the minimum Kirchhoff index among polyomino chains with any prescribed positive number of square cells. The unique minimizing graph is the zigzag chain; its geometric realization is unique up to translation, rotation and reflection. An exact formula requires a linear number of integer-arithmetic steps. The proof embeds the geometric class in a larger family of abstract edge-fused square chains. An electrical recurrence expresses the entire Kirchhoff index as a constant minus a quadratic form in ternary turn variables. The degree restriction forbids two consecutive equal nonzero variables. A positive semidefinite comparison solves this quadratic maximization for every length, and a strict diagonal estimate gives all equality cases. The proof is analytic. Independent exact computations check the network identities, finite geometric enumeration, and finite instances of the matrix comparison.

## 1 Introduction and main result

Replace every edge of a connected finite graph by a resistor of resistance one. Its effective resistance $R_G(x, y)$ accounts for all routes between two vertices, including the reduction in resistance from parallel routes. The Kirchhoff index sums these resistances over unordered vertex pairs:

$$\mathrm{Kf}(G) = \sum_{\{x,y\}\subseteq V(G),\ x\neq y} R_G(x, y). \tag{1}$$

Here successive square cells share an edge, and a shared edge remains one unit resistor throughout.

Sun, Yang and Xu [1] determine the maximum over polyomino chains, with the linear chain as unique maximizer, and study extrema in a restricted bend-free subclass. Their concluding discussion leaves the full-class minimum problem addressed here. Extremal results for $k$-polycyclic chains assume $k \geq 5$ and do not cover square chains with adjacent attachment edges [2]. The similarly named random chains in [3] use cut-edge attachments and degree-weighted indices.

A literature audit on 23 October 2026 found no complete previous solution in the checked sources. This is a bounded search statement, not a guarantee of priority. Relevant definitions and conclusions of the primary paper were read in publisher-indexed text; a complete publisher PDF was not obtained. All electrical and optimization results used below are derived here, without reliance on an inaccessible transformation proof.

**Definition 1.1.** A *polyomino chain with $n$ cells* is the plane graph formed by the sides and vertices of $n$ distinct unit squares of the square lattice, subject to the following conditions:

(i) the graph is 2-connected and its bounded faces are precisely these $n$ constituent square cells;

(ii) each cell has a common side with at most two other cells;

(iii) each vertex belongs to at most three cells.

Write $\mathcal{B}_n$ for this class. When counting geometric realizations, translation and the eight rotations or reflections of the lattice are identified. Unlabelled graph isomorphism is a separate equivalence.

These conditions give the face-based geometric class in [1]. Edge-connected regular unit-square faces have a common orientation and lattice, propagated across shared sides after a rigid motion. There is no extra bounded region or overlapping cell interior. Ordinary corner contact at a turn is allowed. The exclusion of additional remote contact follows in Section 2; it is not assumed from a turn sequence.

Let $L_n$ be the straight row of $n$ cells. Let $Z_n$ be the staircase whose successive cell steps alternate east and north. Set $Z_1 = C_4$; $L_2 = Z_2$ is the domino graph. For $n \geq 2$, the subgraph of $Z_n$ induced by vertices of degree greater than two is a path of $n-1$ edges, matching the source's zigzag characterization.

**Theorem 1.2.** *For every $n \geq 1$ and every $G \in \mathcal{B}_n$,*

$$\mathrm{Kf}(G) \geq \mathrm{Kf}(Z_n),$$

*with equality if and only if $G$ is isomorphic to $Z_n$. Every minimizing geometric realization is a staircase up to translation, rotation and reflection; reversal of cell order produces no additional case.*

*The exact minimum $m_n$ is as follows. Define*

$$q_{-1} = 0, \quad q_0 = 1, \quad q_{k+1} = 4q_k - q_{k-1}, \qquad B_0 = 0, \quad B_k = 2kq_k + q_{k-1} \quad (k \geq 1), \tag{2}$$

*and*

$$P_0 = 0, \qquad P_k = B_k - P_{k-1}, \qquad F_n = \sum_{j=1}^{n-2} B_{n-1-j}(B_j - 2P_{j-1}), \tag{3}$$

*where an empty sum is zero. Then*

$$\boxed{m_n = \frac{(n+1)^2\big((n-1)q_n + q_{n+1}\big) - 3F_n}{3q_n}.} \tag{4}$$

*In particular $m_1 = 5$, $m_2 = 71/5$, and $m_3 = 233/8$.*

The formula uses $O(n)$ arithmetic operations with integers of growing bit length; it requires no optimization or chain enumeration. Figure 1 distinguishes the zigzag from a return bend.

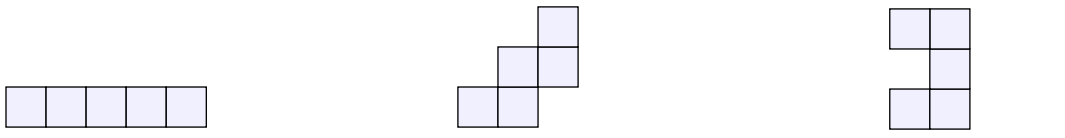



Figure 1: The straight chain, the minimizing zigzag, and a different valid five-cell chain. The last has index $5491/65$, whereas $\mathrm{Kf}(Z_5) = 3205/39$. Every common side is a single graph edge.

## 2 From geometric chains to admissible words

**Lemma 2.1.** *The cell-adjacency graph of a member of $\mathcal{B}_n$ is a path, with the single-vertex convention for $n = 1$. Its graph has $3n+1$ edges and $2n+2$ vertices. In cell-path order, each square after the first meets the preceding graph in exactly its attaching side and adds two new vertices.*

*Proof.* Every vertex is incident to the unbounded face: otherwise all four lattice sectors around it would be occupied by bounded square faces, contrary to condition (iii). A 2-connected plane graph has a simple outer-face boundary, so the graph is outerplane. Its bounded-face dual is a tree: the interior edges are noncrossing chords of the outer boundary, and insertion of each chord increases both the number of regions and the number of dual edges by one. The dual remains connected and is therefore a tree. This dual has maximum degree at most two by condition (ii), and is therefore a path.

There are $n-1$ shared sides. Counting each once gives $E = 4n-(n-1) = 3n+1$. Euler's formula gives $V = E-n+1 = 2n+2$. Order the cells from an end of their path. Each new square shares its entering side with its predecessor and adds at most two vertices. The final upper bound is $4+2(n-1) = 2n+2$, and equality holds. Therefore every attachment adds two new vertices; no other identification with an earlier vertex is possible. A corner may already belong to the previous two cells, but it is then an endpoint of the attaching side. □

For an *abstract extension*, start with $C_4$ and an oriented edge $(u, v)$. Keep its unit edge $uv$, introduce fresh vertices $a, b$, and add the unit path $u-a-b-v$. For a following extension, choose the next oriented boundary by

$$c=-1:\ (u,a), \qquad c=0:\ (a,b), \qquad c=1:\ (b,v). \tag{5}$$

For $n \geq 2$ cells the relevant word has $n-2$ entries: $c_i$ is selected after attaching cell $i+1$ and specifies where cell $i+2$ attaches. The exit after the final square has no effect on the graph. Lemma 2.1 represents every geometric chain in this way.

**Lemma 2.2.** *For $n \geq 2$, an abstract extension word has maximum degree at most four if and only if no two adjacent entries are equal and nonzero. Hence every geometric chain yields a word in*

$$\mathcal{C}_m = \{c \in \{-1,0,1\}^m : (c_i, c_{i+1}) \notin \{(1,1),(-1,-1)\}\}, \qquad m = n-2. \tag{6}$$

*Proof.* A fresh vertex starts with degree two; each attachment raises the degree of each entering-edge endpoint by one. Once omitted from an exit edge, a vertex cannot return to a later boundary, since subsequent vertices are fresh. A first endpoint persists only under $c=-1$, and a second only under $c=1$. Two consecutive equal nonzero choices retain the same endpoint through three attachments, giving degree at least five. Conversely, degree five requires this persistence through three attachments, and hence a forbidden pair. At most three incident cells in the geometric class imply maximum degree four. □

The abstract family is a strict relaxation. The seven-cell word $(-1,0,-1,0,-1)$ satisfies (6), but its forced lattice placement makes its last cell touch the first at an additional vertex. The abstract construction keeps these vertices distinct. No assertion that all allowed words are geometrically realizable is used. A lower bound over the relaxed class applies to geometric chains; only its equality words must be realized.

# 3 An exact electrical recurrence

## 3.1 Complete boundary data and all pair contributions

For the Laplacian $L$, solving the unit-current equation gives

$$R_G(x,y) = (e_x-e_y)^T L^+ (e_x-e_y), \qquad L^+\mathbf{1} = 0.$$

The same formula uses a grounded Laplacian inverse with grounded coordinate zero. Summing over unordered pairs yields

$$\mathrm{Kf}(G) = |V(G)|\,\mathrm{tr}(L^{+}). \tag{7}$$

For a graph with $N$ vertices and oriented unit edge $(u, v)$, set

$$\begin{aligned} r &= R_G(u,v), & d_x &= R_G(x,u) - R_G(x,v),\\ D &= \sum_x d_x, & Q &= \sum_x d_x^2,\\ T &= \sum_x \big(R_G(x,u) + R_G(x,v)\big), & K &= \mathrm{Kf}(G). \end{aligned} \tag{8}$$

All sums include the endpoints. In this section $T$ is a scalar; the inverse matrix denoted $T$ later is defined separately.

**Proposition 3.1.** *After extension and choice* (5)*, put $h = r + 3$, $A = N - r - 1$. The state obeys*

$$N' = N + 2, \qquad r' = \frac{r+2}{h}, \tag{9}$$

$$D' = \frac{D + 2Ac}{h}, \qquad Q' = \frac{Q + 4cD + 4Ac^2 + 2(r+2)^2}{h^2}, \tag{10}$$

$$T' = T + \frac{-(Q + Nr^2)/2 + 4N + 2r + 4 - 2cD - 2Ac^2}{h}, \tag{11}$$

$$K' = K + T - \frac{(N+2)Q - D^2 + 2Nr^2 - 16N - 4r - 8}{4h}. \tag{12}$$

*The initial square has*

$$(N, r, T, D, Q, K) = (4, 3/4, 5, 0, 5/4, 5). \tag{13}$$

*Proof.* Eliminating $a, b$ adds conductance $1/3$ between old vertices $u, v$. Write $R_*$ for the reduced resistances. A rank-one inverse update of a grounded Laplacian gives

$$R_*(x,y) = R_G(x,y) - \frac{(d_x - d_y)^2}{4h}. \tag{14}$$

The relevant bilinear form is $(e_x - e_y)^T L^{+}(e_u - e_v) = (d_y - d_x)/2$, and the update denominator is $3 + r$. In particular $R_*(u,v) = 3r/h$.

For the new vertices, the internal path block and its inverse are

$$C = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Block inversion gives harmonic endpoint weights $(2/3, 1/3)$ at $a$ and $(1/3, 2/3)$ at $b$, and contributes $2/3$ to each internal diagonal. Thus, for $z_t = a, b$ at $t = 1/3, 2/3$ respectively,

$$R(z_t, x) = (1-t)R_*(u,x) + tR_*(v,x) + t(1-t)\big(3 - R_*(u,v)\big). \tag{15}$$

This is simply expansion of the resistance quadratic form in the block inverse, so no probabilistic assumption is involved. Substitution gives

$$R(a,x) = \frac{2R_G(u,x) + R_G(v,x)}{3} - \frac{d_x^2 + (2r/3)d_x + r^2 - 8}{4h}, \tag{16}$$

$$R(b,x) = \frac{R_G(u,x) + 2R_G(v,x)}{3} - \frac{d_x^2 - (2r/3)d_x + r^2 - 8}{4h}. \tag{17}$$

Every new edge has resistance $(r+2)/(r+3)$, being a unit edge parallel to complementary resistance $r+2$.

The contributions to the new Kirchhoff index from old–old, old–new, and new–new pairs are respectively

$$K - \frac{NQ - D^2}{4h}, \qquad T - \frac{Q + Nr^2 - 8N}{2h}, \qquad \frac{r+2}{h}.$$

The first uses $\sum_{x<y}(d_x - d_y)^2 = NQ - D^2$; the second sums (16) and (17). Their sum proves (12), with every unordered pair included once.

For an old vertex, its new boundary difference is $(d_x + 2c)/h$. The two new differences are

| $c$ | $d'_a$ | $d'_b$ |
|---|---|---|
| $-1$ | $(r+2)/h$ | $r/h$ |
| $0$ | $-(r+2)/h$ | $(r+2)/h$ |
| $1$ | $-r/h$ | $-(r+2)/h$ |

Their first and second moments prove (10). Writing $S'_w = \sum_x R(w,x)$ in the extended graph, the same identities give

$$\begin{aligned}
S'_u &= T/2 + 3D/(2h) - (Q + Nr^2 - 12r - 16)/(4h),\\
S'_v &= T/2 - 3D/(2h) - (Q + Nr^2 - 12r - 16)/(4h),\\
S'_a &= T/2 + D/(2h) - (Q + Nr^2 - 8N - 4r - 8)/(4h),\\
S'_b &= T/2 - D/(2h) - (Q + Nr^2 - 8N - 4r - 8)/(4h).
\end{aligned}$$

Adding the chosen two sums proves (11). The initial values follow from adjacent and opposite resistances $3/4$ and $1$ in $C_4$. □

### 3.2 The quadratic objective and its coefficient telescope

Let $K_n(c)$ be the index of an abstract word and $K_n(0)$ that of the straight word. No degree or geometric restriction is needed next.

**Proposition 3.2.** *For $n \geq 3$, put $m = n-2$ and*

$$H_{ij} = B_{\min(i,j)} B_{n-1-\max(i,j)} \quad (1 \leq i, j \leq m). \tag{18}$$

*Then every abstract word satisfies*

$$\boxed{K_n(c) = K_n(0) - \frac{c^T H c}{q_n}.} \tag{19}$$

*Proof.* At $k$ cells, induction in the state recurrence gives

$$N_k = 2k+2, \quad r_k = 1 - \frac{q_{k-1}}{q_k}, \quad h_k = \frac{q_{k+1}}{q_k}, \quad A_k = \frac{B_k}{q_k}. \tag{20}$$

Put $E_k = q_k D_k$, $U_k = q_k^2 Q_k$, and let $\Delta$ subtract the straight-word value. The moment recurrences give

$$E_k = 2\sum_{i<k} B_i c_i, \tag{21}$$

$$\Delta U_k = 4\sum_{i<k} q_i B_i c_i^2 + 8 \sum_{i<j<k} B_i q_j c_i c_j. \tag{22}$$

Indeed $E_{k+1} - E_k = 2B_k c_k$ and $\Delta U_{k+1} - \Delta U_k = 4q_k c_k E_k + 4q_k B_k c_k^2$.

Rewrite the scalar boundary update as

$$T_{k+1} = T_k - \frac{h_k Q_{k+1}}{2} + C_k, \qquad C_k = \frac{-N_k r_k^2/2 + 4N_k + 2r_k + 4 + (r_k+2)^2}{h_k}.$$

Here $C_k$ depends only on $k$. Hence $\Delta T_k = -\sum_{\ell=2}^{k} \Delta U_\ell/(2q_{\ell-1}q_\ell)$. Subtracting straight values in (12) and summing yields

$$\Delta K_n = \sum_{k=2}^{n-1} \bigl(-\alpha_k \Delta U_k + \beta_k E_k^2\bigr), \tag{23}$$

where

$$\alpha_k = \frac{n-k}{2q_{k-1}q_k} + \frac{k+2}{2q_k q_{k+1}}, \qquad \beta_k = \frac{1}{4q_k q_{k+1}}.$$

After (21)–(22) are substituted, the negative coefficient of $c_i^2$ is $B_i C_i^{(n)}$, and of $c_i c_j$ for $i < j$ is $2B_i C_j^{(n)}$, with

$$C_j^{(n)} = \sum_{k=j+1}^{n-1} \left\{ \frac{2q_j(n-k)}{q_{k-1}q_k} + \frac{2q_j(k+2) - B_j}{q_k q_{k+1}} \right\}. \tag{24}$$

We prove $C_j^{(n)} = B_{n-1-j}/q_n$ explicitly. The recurrence gives

$$q_s q_t - q_{s+1}q_{t-1} = q_{s-t} \quad (s \geq t \geq 1), \tag{25}$$

$$\sum_{k=j+1}^{n} \frac{1}{q_{k-1}q_k} = \frac{q_{n-j-1}}{q_j q_n}. \tag{26}$$

The first is an induction; for the second, take the difference of the right side at $n+1$ and $n$ and apply the first, starting at $n = j+1$.

For $n = j+2$, (24) equals $9/q_{j+2} = B_1/q_{j+2}$. Increasing $n$ by one changes that sum by

$$\frac{2q_{n-j-1}}{q_n} + \frac{2q_j(n+2) - B_j}{q_n q_{n+1}}. \tag{27}$$

Put $t = n-j-1$. Subtract $2q_t/q_n$ from $B_{t+1}/q_{n+1} - B_t/q_n$. The resulting numerator over $q_n q_{n+1}$ is

$$\begin{aligned} &2(t+1)(q_{t+1}q_n - q_t q_{n+1}) + (q_t q_n - q_{t-1}q_{n+1}) \\ &\quad = 2(t+1)q_j + q_{j+1} = 2q_j(n+2) - B_j. \end{aligned}$$

Here (25) and $q_{j+1} = 4q_j - q_{j-1}$ were used. This matches (27), completing the induction. The coefficients in (23) are therefore exactly those of $H/q_n$. □

## 4 A strict estimate for the Green kernel

We next analyze (18) independently of the network recurrence. Put $N = n-1$, $m = N-1$, and

$$v_i = (-1)^i, \qquad V = \operatorname{diag}(v), \qquad h = VHv.$$

Here $h$ is a vector, distinct from the scalar extension denominator used in the previous section.

**Proposition 4.1.** *The matrix $H$ is positive definite. Its inverse $T = H^{-1}$ is tridiagonal with negative adjacent off-diagonal entries. Each $h_i$ is positive, and*

$$T_{ii} h_i > \frac{2}{3} \qquad (1 \leq i \leq m). \tag{28}$$

### 4.1 The inverse and the alternating response

Set $b_i = B_{N-i}$ and $R_i = B_i/b_i$. The $B_i$ are positive and strictly increasing, so $R_i$ is strictly increasing. Then

$$H = \operatorname{diag}(b_i) C \operatorname{diag}(b_i), \qquad C_{ij} = \min(R_i, R_j).$$

With $R_0 = 0$ and $\delta_i = R_i - R_{i-1} > 0$, the factorization

$$C = U \operatorname{diag}(\delta_1, \ldots, \delta_m) U^T, \qquad U_{ij} = \begin{cases} 1 & j \le i, \\ 0 & j > i \end{cases}$$

proves positive definiteness. The inverse of $U$ is the first-difference matrix, giving

$$T_{ii} = \frac{1}{b_i^2}\left(\frac{1}{\delta_i} + \frac{1}{\delta_{i+1}}\right), \qquad T_{i,i+1} = -\frac{1}{b_i b_{i+1} \delta_{i+1}}. \tag{29}$$

At $i = m$, $1/\delta_{m+1}$ means zero, solely as a convention for the absent neighbor.

The recurrence $P_k = B_k - P_{k-1}$ expresses $P_k = \sum_{j=1}^{k} (-1)^{k-j} B_j$. Substitution into that recurrence proves

$$P_k = \frac{(10k+3)q_k + (1-2k)q_{k-1} - 3(-1)^k}{6}. \tag{30}$$

For $p = i$ and $r = N - i$, splitting the alternating row sum at $i$ yields

$$\frac{h_i}{B_p B_r} = 1 - x_p - x_r, \qquad x_k = \frac{P_{k-1}}{B_k}. \tag{31}$$

Define $a_k = B_{k-1}/B_k$, so $a_1 = 0$. Multiplying the diagonal formula (29) by $H_{ii}$ gives

$$T_{ii} H_{ii} = \frac{1}{1 - a_p a_{r+1}} + \frac{1}{1 - a_{p+1} a_r} - 1. \tag{32}$$

Indeed the two unsimplified terms are $R_i/(R_i - R_{i-1})$ and $R_i/(R_{i+1} - R_i)$; their relevant ratios are $a_p a_{r+1}$ and $a_{p+1} a_r$. At the last index $r = 1$, the second term is zero since $a_1 = 0$. The formula thus includes both endpoints without division by $B_0$.

### 4.2 Elementary scalar bounds

Put

$$\rho = 2 - \sqrt{3}, \qquad \alpha = 2 + \sqrt{3} = \rho^{-1}, \qquad s = \frac{\rho}{1+\rho}.$$

The recurrence and initial values imply

$$q_k - \alpha q_{k-1} = \rho^k, \qquad \rho q_k - q_{k-1} = \rho^{k+1}. \tag{33}$$

Thus $0 \le a_k < \rho$, because

$$B_k - \alpha B_{k-1} = 2\alpha q_{k-1} + \rho^{k-1} + 2k\rho^k > 0.$$

**Lemma 4.2.** *For every $k \ge 1$,*

$$a_k \ge \rho\left(1 - \frac{1}{k}\right), \qquad x_k < s - \frac{1}{8k}. \tag{34}$$

*Proof.* Expansion using (33) gives

$$kB_{k-1} - (k-1)\rho B_k = \rho\big(q_{k-1} - k\rho^{k-1} - 2k(k-1)\rho^k\big).$$

The integer sequence satisfies $q_{k-1} \geq k$. For $t = k-1 \geq 0$, Bernoulli's inequality and $\alpha - 1 > 2\rho$ give $\alpha^t \geq 1 + t(\alpha-1) \geq 1 + 2t\rho$. Hence $\rho^t(1+2t\rho) \leq 1$, proving the first bound, including equality at $k = 1$.

For the second, (30) simplifies to

$$s - x_k = \frac{2q_k + (1+2k\rho)\rho^k - 3(-1)^k}{6B_k}. \tag{35}$$

After multiplication by positive denominators, the desired bound is equivalent to

$$2kq_k - 3q_{k-1} + 4k(1+2k\rho)\rho^k - 12k(-1)^k > 0. \tag{36}$$

For $k \geq 2$, the recurrence gives $q_k \geq 3q_{k-1}$ and $q_{k-1} \geq 4$. The first, second and last terms together are at least $(6k-3)4 - 12k = 12k - 12 > 0$; the remaining term is positive. For $k = 1$, use $x_1 = 0$ and $s > 1/8$ directly. □

*Completion of Proposition 4.1.* Set $u = 1/p + 1/r$, so $0 < u \leq 2$. The second scalar bound gives

$$1 - x_p - x_r > \frac{1}{\sqrt{3}} + \frac{u}{8} > 0. \tag{37}$$

In particular $h_i > 0$. The convex function $f(z) = z/(1-z)$ is bounded below by its tangent at $\rho^2$. The first scalar bound implies

$$\begin{aligned} a_p a_{r+1} + a_{p+1} a_r - 2\rho^2 &\geq -\rho^2 \left(\frac{1}{p} + \frac{1}{r+1} + \frac{1}{p+1} + \frac{1}{r}\right) \\ &\geq -2\rho^2 u. \end{aligned}$$

Since $1 + 2f(\rho^2) = 2/\sqrt{3}$ and $2\rho^2 f'(\rho^2) = 1/6$, equation (32) gives

$$T_{ii} H_{ii} \geq \frac{2}{\sqrt{3}} - \frac{u}{6} > 0. \tag{38}$$

Both lower bounds are positive, so multiplication gives

$$\begin{aligned} T_{ii} h_i &> \left(\frac{1}{\sqrt{3}} + \frac{u}{8}\right)\left(\frac{2}{\sqrt{3}} - \frac{u}{6}\right) \\ &= \frac{2}{3} + \frac{u}{12\sqrt{3}} - \frac{u^2}{48} > \frac{2}{3}, \end{aligned}$$

because $u \leq 2 < 4/\sqrt{3}$. The estimate is strict for all finite sizes. □

# 5 Global quadratic optimization and equality

**Theorem 5.1.** *For $H$ in (18) and every $c \in \mathcal{C}_m$,*

$$c^T H c \leq v^T H v, \qquad v_i = (-1)^i,$$

*with equality exactly for $c = v$ and $c = -v$.*

*Proof.* Use $T = H^{-1}$, $h = VHv$ and $D = \mathrm{diag}(h)$. The matrix $T' = VTV$ is positive definite and entrywise nonnegative: its adjacent off-diagonal entries are positive and all more distant ones vanish. Its definition gives

$$T'h = \mathbf{1}. \tag{39}$$

Therefore $T'D$ is a nonnegative row-stochastic matrix, similar to the symmetric positive definite matrix

$$W = D^{1/2}T'D^{1/2}.$$

The diagonal in row $i$ is $d_i = T_{ii}h_i > 2/3$; the remaining row sum is $1 - d_i$. Gershgorin's theorem puts every eigenvalue strictly above $1/3$. Row stochasticity bounds all eigenvalues in absolute value by one and supplies the eigenvalue one. The spectrum of $W$ lies in $(1/3, 1]$.

On this interval,

$$(4 - 3x) - x^{-1} = \frac{(1-x)(3x-1)}{x} \geq 0.$$

Applying this inequality in an orthonormal eigenbasis gives $W^{-1} \preceq 4I - 3W$, where $\preceq$ denotes the positive semidefinite order. Congruence by $D^{1/2}$ and then by $V$ yields

$$H \preceq J := 4D - 3DTD. \tag{40}$$

Since $Dv = Hv$, we have $TDv = v$ and consequently

$$Jv = Hv. \tag{41}$$

Define strictly positive coefficients

$$\lambda_i = -6h_iT_{i,i+1}h_{i+1} \quad (1 \leq i < m), \qquad \mu_i = h_i(3T_{ii}h_i - 2) \quad (1 \leq i \leq m). \tag{42}$$

Their signs follow from Proposition 4.1. The tridiagonal form and (39) give the exact identity

$$c^TJc = \sum_{i=1}^{m-1} \lambda_i(c_i^2 + c_{i+1}^2 + c_ic_{i+1}) + \sum_{i=1}^{m} \mu_ic_i^2. \tag{43}$$

In detail, the cross coefficient is $2J_{i,i+1} = -6h_iT_{i,i+1}h_{i+1} = \lambda_i$. The adjacent $\lambda$ terms contribute $6h_i(1 - T_{ii}h_i)$ to the coefficient of $c_i^2$, by (39). Subtracting that from $J_{ii} = 4h_i - 3T_{ii}h_i^2$ leaves $\mu_i$. Absent endpoint terms are zero.

Every allowed ternary pair satisfies

$$c_i^2 + c_{i+1}^2 + c_ic_{i+1} \leq 1. \tag{44}$$

This follows by inspecting $(0,0)$, $(\pm 1, 0)$, $(0, \pm 1)$, $(1, -1)$ and $(-1, 1)$. Also $c_i^2 \leq 1$. The alternating vector saturates them all. Equations (40)–(43) now give

$$c^THc \leq c^TJc \leq \sum_i \lambda_i + \sum_i \mu_i = v^TJv = v^THv. \tag{45}$$

Equality forces $c_i^2 = 1$ for every $i$, since every $\mu_i$ is strictly positive. Admissibility then forces $c = v$ or $c = -v$; both attain equality. For $m = 1$, $H = [81]$, $W = [1]$, and $\mu_1 = 81 > 0$, with no pair terms. Thus the smallest matrix size introduces no additional equality case. □

## 6 Attainment, exact values, and small cases

**Proposition 6.1** (Recovery of the known maximum)**.** *For every $n \geq 1$ and $G \in \mathcal{B}_n$,* $\mathrm{Kf}(G) \leq \mathrm{Kf}(L_n)$*, with equality exactly for the linear chain.*

*Proof.* For $n \geq 3$, positive definiteness of $H$ and (19) give $K_n(c) \leq K_n(0)$, with equality exactly when $c = 0$. Every zero choice takes the opposite side of the current square, so the geometric chain is a straight row. For $n = 1, 2$ there is only one chain. This recovers the maximum theorem of [1] directly from the same objective formula. □

*Proof of Theorem 1.2.* For $n \geq 3$, Lemmas 2.1 and 2.2 give an allowed word for every geometric chain. Since $q_n > 0$, the quadratic identity and Theorem 5.1 imply

$$\mathrm{Kf}(G) \geq K_n(0) - \frac{v^T H v}{q_n},$$

and equality forces an alternating word.

Fix the first two cell coordinates as $(0,0),(1,0)$. One alternating choice has successive steps east, north, east, north, and so on; the other is its reflection. These staircases are valid for every $n$. Gluing cycles along edges preserves 2-connectivity. Each step raises a fixed linear coordinate by one. Cells separated by at least three positions have center Manhattan distance at least three and cannot share a side or corner. Cells two positions apart meet only at the allowed corner of the intervening cell. Hence there are no additional contacts, all bounded faces are the cells, and at most three cells meet at a vertex.

The successive common sides form a path of $n-1$ edges. Its endpoints have degree three, its internal vertices degree four, and all remaining vertices degree two. Monotonicity prevents an extra edge between nonconsecutive path vertices. The graph is therefore $Z_n$. Once the first two cells and orientation are fixed, a turn word determines all remaining cell positions. Its two equality words are reflections, giving one geometric and one graph-isomorphism class. This reasoning applies to every minimizing embedding; it does not assume general uniqueness of lattice embeddings of an abstract graph.

The alternating value reduces to a single sum:

$$\begin{aligned} v^T H v &= \sum_{j=1}^{n-2} B_{n-1-j} \left( B_j + 2 \sum_{i<j} (-1)^{j-i} B_i \right) \\ &= \sum_{j=1}^{n-2} B_{n-1-j}(B_j - 2P_{j-1}) = F_n. \end{aligned}$$

Appendix A derives

$$K_n(0) = \frac{(n+1)^2((n-1)q_n + q_{n+1})}{3q_n}. \tag{46}$$

This proves (4).

For $n = 1$ the only graph is $C_4$, with index $4(3/4) + 2 = 5$. For $n = 2$ the only graph is the domino, with index $71/5$ from the recurrence. Both agree with the empty-sum convention. For $n = 3$, the unique minimizing graph is the L triomino, with $233/8$; the straight triomino has $214/7$. All sizes and equality cases are covered. □

| Cells $n$ | Geometric chains up to lattice symmetry | Exact minimum $m_n$ |
|---|---|---|
| 1 | 1 | 5 |
| 2 | 1 | 71/5 |
| 3 | 2 | 233/8 |
| 4 | 3 | 565/11 |
| 5 | 7 | 3205/39 |
| 6 | 13 | 357831/2911 |
| 7 | 30 | 1900383/10864 |
| 8 | 64 | 9711229/40545 |
| 9 | 150 | 12030917/37829 |
| 10 | 338 | 232556377/564719 |

Table 1: Small exact values. Every row has one minimizing geometric class and one minimizing graph-isomorphism class. The counts come from finite enumeration; the general minimum and equality classification follow from the analytic proof.

# 7 Reproducibility and verification scope

The universal theorem depends on no computational reduction or finite search assertion. The package's exact computations check intermediate identities and consequences independently. Floating-point ties are not used to classify minimizers.

**Electrical and matrix checks.** The standard-library program `src/recurrence_check.py` constructs abstract graphs and compares every boundary-state component against exact grounded-Laplacian inversion. Through five cells this checks 121 extension words, including the initial square. The program `src/verify_theorem.py` checks the quadratic formula on all 1,092 words through eight cells, including degree-violating words, and classifies the allowed-word minima exactly. For $3 \le n \le 24$ it also constructs the matrix comparison and checks inverse identities, positive coefficients, contact with the alternating vector, and positive semidefiniteness by rational elimination. These finite checks support the analytic proof; they do not replace it.

**Geometric enumeration.** The generator `src/exact_chains.py` grows a chain at either end. A proposed square must be new, have exactly one side-neighbor in the prefix, and have no vertex intersection beyond the entering side. Deleting a terminal cell and applying Lemma 2.1 proves this generation complete. Translation and all eight signed coordinate permutations produce a canonical geometric form. Cells are unlabelled, so chain reversal produces no duplicate. Shared sides are stored as a single unordered edge. Rational inversion computes the index; graph isomorphism separately groups geometric outputs. The recorded enumeration extends through fifteen cells. For $n = 11, 12, 13, 14, 15$ its respective counts are $794, 1836, 4313, 10067, 23621$, with a single zigzag minimum in every case.

The independent `src/enumeration_audit.py` instead grows all connected free polyominoes, then filters completed shapes by their cell dual, incidence counts, 2-connectivity, and $E - V + 1 = n$. Through eight cells its accepted sets agree exactly with the terminal-extension generator. For all fourteen accepted shapes through five cells, a separate full-Laplacian construction computes

$$L^+ = (L + \mathbf{1}\mathbf{1}^T/N)^{-1} - \mathbf{1}\mathbf{1}^T/N$$

with exact rational arithmetic and agrees with the grounded calculation. This comparison changes both the generation procedure and the Laplacian representation. Merely re-summing resistances from one inverse would be a weaker check.

**Reproduction.** From the root of the accompanying package, run

```
python src/minimum.py 10
python src/recurrence_check.py --max-cells 5
python src/verify_theorem.py --max-n 24 --word-n 8 --laplacian-n 5
python src/enumeration_audit.py --max-n 8
```

The first three core computations use Python's standard library. The geometric audit additionally uses `python-flint`, `SymPy` and `NetworkX`. The README gives installation and complete enumeration commands; the package includes recorded data. Every essential mathematical claim, including generation completeness, is explained in this paper without dependence on unpublished notes.

**Scope and contributions.** The full maximum theorem and restricted-subclass analysis are prior work. This manuscript gives the full minimum through the abstract-word quadratic formula and strict spectral comparison, including all equality cases and the value formula. The dated search found no prior full solution but cannot exclude unlocated work. The publisher-indexed Theorem 5.6 has an internal minimum/maximum wording inconsistency, and the cell-order description has an apparent index-range typo. Neither inconsistent statement is a premise of this proof.

The derivations, manuscript, code and internal adversarial checks were produced by GPT-6 Astra, an AI system. No Lean or other proof-assistant formalization was compiled, and no external human peer review is claimed. Verification consists of the explicit analytic proof, internal mathematical review and the specified exact executable checks.

# A The straight-chain value from a path determinant

Put $s = n + 1$. The straight chain is the Cartesian product $P_s \square P_2$. If $0 = \lambda_0 < \lambda_1 \le \cdots \le \lambda_{s-1}$ are the path Laplacian eigenvalues, the product Laplacian is a Kronecker sum, with eigenvalues $\lambda_k$ and $\lambda_k + 2$. Equation (7) gives

$$K_n(0) = 2s \left( \sum_{k=1}^{s-1} \frac{1}{\lambda_k} + \sum_{k=0}^{s-1} \frac{1}{\lambda_k + 2} \right). \tag{47}$$

In a path, resistance equals graph distance, so

$$\mathrm{Kf}(P_s) = \sum_{d=1}^{s-1} d(s-d) = \frac{s(s^2-1)}{6}.$$

Thus the first sum in (47) is $(s^2 - 1)/6$.

Define polynomials $Q_{-1}(x) = 0$, $Q_0(x) = 1$ and $Q_{k+1}(x) = xQ_k(x) - Q_{k-1}(x)$, so $Q_k(4) = q_k$. For $s \ge 2$, $tI + L(P_s)$ is tridiagonal with endpoint diagonal entries $t + 1$, interior entries $t + 2$, and off-diagonal entries $-1$. Set $x = t + 2$. Before the last endpoint, its leading minors satisfy $D_k = Q_k(x) - Q_{k-1}(x)$, by the initial cases and $D_k = xD_{k-1} - D_{k-2}$. Including the last endpoint gives

$$\begin{aligned} \det(tI + L(P_s)) &= (x-1)(Q_{s-1} - Q_{s-2}) - (Q_{s-2} - Q_{s-3}) \\ &= (x-2)Q_{s-1}(x) = tQ_{s-1}(t+2). \end{aligned}$$

For $s = 2$ the same identity follows directly from the two-by-two determinant; for $s = 1$ both sides are $t$. Its logarithmic derivative at $t = 2$ is

$$\sum_{k=0}^{s-1} \frac{1}{\lambda_k + 2} = \frac{1}{2} + \frac{Q_n'(4)}{q_n}.$$

Differentiating the recurrence gives

$$Q_n'(4) = \frac{(n+1)(q_{n+1} - q_{n-1}) - 4q_n}{12}. \tag{48}$$

For an explicit induction check, the right side is zero at $n = 0$ and one at $n = 1$, and obeys $Q_{n+1}'(4) = q_n + 4Q_n'(4) - Q_{n-1}'(4)$. Substituting into (47) yields

$$K_n(0) = \frac{(n+1)^3}{3} + \frac{(n+1)^2(q_{n+1} - q_{n-1})}{6q_n} = \frac{(n+1)^2((n-1)q_n + q_{n+1})}{3q_n},$$

using $q_{n-1} = 4q_n - q_{n+1}$. This is (46).

## References


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